**Project Design Phase-II**
**Technology Stack (Architecture & Stack)**

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|---|---|
| Date | 28 September 2026 |
| Team ID | SWTID-2026-2598 |
| Project Name | AI BlogNest API |
| Maximum Marks | 4 Marks |

**Technical Architecture**

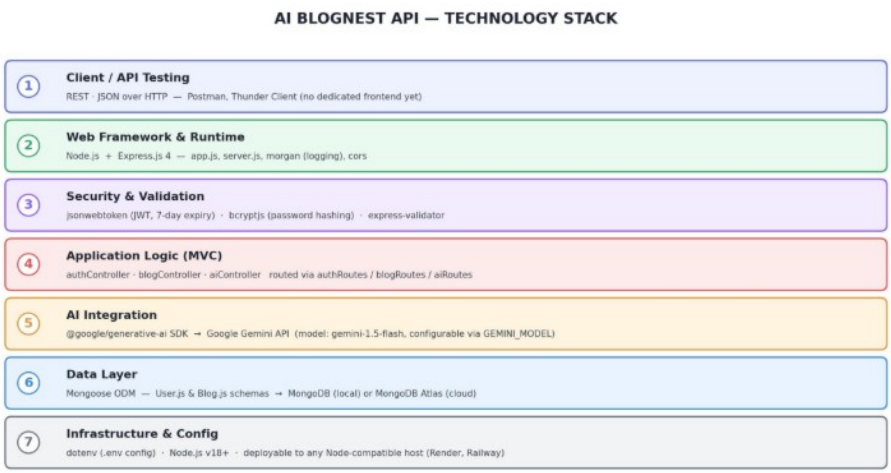



**Table-1: Components & Technologies**

| S.No. | Component | Description | Technology |
|---|---|---|---|
| 1. | User Interface | How a client interacts with the application — currently API-only, tested via Postman / Thunder Client. | REST / JSON over HTTP |
| 2. | Application Logic-1 (Auth) | Registration, login, and route protection. | jsonwebtoken, bcryptjs |
| 3. | Application Logic-2 (Blog Management) | Blog CRUD logic and ownership enforcement. | Node.js, Express.js |
| 4. | Application Logic-3 (AI Orchestration) | Builds prompts, calls Gemini, and cleans up generated text. | @google/generative-ai (Gemini) |
| 5. | Database | Stores Users and Blogs collections. | MongoDB + Mongoose ODM |
| 6. | Cloud Database | Optional managed database hosting. | MongoDB Atlas |

| S.No. | Component | Description | Technology |
|---|---|---|---|
| 7. | File Storage | Not implemented in the current version (no file/image uploads). | N/A |
| 8. | External API-1 | AI-powered blog content generation and summarization. | Google Gemini API |
| 9. | External API-2 | Not implemented in the current version. | N/A |
| 10. | Machine Learning Model | Large language model used for text generation, accessed via API (not self-hosted). | gemini-1.5-flash (default) |
| 11. | Infrastructure (Server / Cloud) | Local Node.js server for development; deployable to any Node-compatible host. Local Server Configuration: Node.js v18+, 8 GB RAM. Cloud Server Configuration: Not yet deployed. | Local / Cloud (e.g. Render, Railway) + MongoDB Atlas |

**Table-2: Application Characteristics**

| S.No. | Characteristics | Description | Technology |
|---|---|---|---|
| 1. | Open-Source Frameworks | Web framework and ODM used to build the API. | Express.js, Mongoose |
| 2. | Security Implementations | Password hashing, token-based auth, input validation, and centralized error handling. | bcrypt, JWT, express-validator |
| 3. | Scalable Architecture | Stateless 3-tier REST architecture (routes → controllers → models); Express instances can scale horizontally. | Node.js, Express.js |
| 4. | Availability | Can run under a process manager for restart-on-crash; MongoDB Atlas offers managed uptime. | PM2 (optional), MongoDB Atlas |
| 5. | Performance | Non-blocking async I/O | Node.js event loop, |

| S.No. | Characteristics | Description | Technology |
|---|---|---|---|
| | | for all routes; JSON payloads kept small; latency is dominated by the external Gemini call, not the database. | MongoDB |